8. VOYAGER 1

Mission title: THE FARTHEST TRAVELER

FIELD	SPECIFICATION
Primary objective	Fly Voyager 1 past Jupiter and Saturn, capture a view of Earth from far away, and detect the crossing into interstellar space.
Mission setting	Deep space: Jupiter, Io, Saturn and Titan flybys, then the outer edge of the Sun's influence. There is no Moon or Mars surface in this mission.
Gameplay type	Spacecraft navigation, camera aiming, instrument reading, and antenna alignment. The spacecraft never lands.

8.1 REAL MISSION CONTEXT

Voyager 1 is a NASA space probe launched on September 5, 1977. It flew past Jupiter in March 1979 and Saturn in November 1980, returning close-up images and measurements of the planets, their rings and their moons, including Io's active volcanoes and Titan's thick atmosphere. In 1990 it took the "Pale Blue Dot" image of Earth from billions of kilometers away. In August 2012 it crossed the heliopause and became the first human-made object to enter interstellar space. It is powered by radioisotope generators, carries a Golden Record of sounds and images from Earth, and still communicates through NASA's Deep Space Network.

8.2 REQUIRED GAMEPLAY SEQUENCE

PHASE / TRIGGER	PLAYER SEES	PLAYER ACTION	GAME RESPONSE / NEXT STATE
1. Mission briefing	A Voyager 1 diagram showing the antenna, cameras, instruments and Golden Record, plus three objectives.	Press "Start Mission".	Begin the launch cinematic.
2. Launch	Rocket on the launch pad at Cape Canaveral. A "Launch Complete" message appears after separation.	No control required; allow "Skip".	Show Voyager 1 alone in space, folded for launch, with an "Unfold Instruments" button.
3. Unfold instruments	Voyager 1 with its boom and dish antenna still folded.	Press "Unfold Instruments".	Animate the boom and antenna opening. Show a short cruise transition toward Jupiter.
4. Jupiter flyby	Jupiter and Io, a camera crosshair, and a highlighted target on Io.	Use directional controls to line the crosshair up with the target, then press "Capture Image".	Show a simulated camera frame of a volcanic plume on Io, labelled as illustrative. Mark objective 1 complete.
5. Gravity assist	A path line curving around Jupiter toward Saturn.	Press "Use Gravity Assist".	Animate Voyager 1 swinging past Jupiter and gaining speed. Do not show the player firing engines for this boost.
6. Saturn and Titan	Saturn with its rings, and Titan with a hazy atmosphere marker.	Move the crosshair to Titan and press "Study Atmosphere".	Show a short card about Titan's thick haze. The path then bends away from the planets toward deep space.
7. Look back at Earth	A "Turn Camera Toward Earth" prompt. Earth is a tiny point of light.	Press "Turn Camera Toward Earth", then "Capture Image".	Show Earth as a tiny pale dot in a band of scattered sunlight, labelled "Pale Blue Dot, 1990". Mark objective 2 complete.
8. Reach interstellar space	A plasma-wave instrument panel and a simple density graph.	Press "Listen to Plasma Waves".	The graph rises sharply and "Heliopause Crossed" appears. Mark objective 3 complete.
9. Send signal home	An antenna alignment guide, an Earth marker, and a Deep Space Network dish inset.	Align the antenna with Earth and press "Send Data".	Animate a signal traveling to Earth with a long delay indicator.
10. Learning result	A Golden Record card and a short note on how far Voyager 1 has traveled.	Press "Continue".	Open completion screen.

8.3 BUTTON AND STATE RULES

- **Unfold Instruments** appears only after the launch is complete.
- **Capture Image** appears at Jupiter only when the crosshair is on the highlighted target.

- **Study Atmosphere** appears only when the crosshair is on Titan.
- **Turn Camera Toward Earth** appears only after the Saturn flyby is complete.
- **Listen to Plasma Waves** appears only after the Earth image has been captured.
- **Send Data** appears only when the antenna is aligned with Earth.
- Voyager 1 cannot land or return. Do not add a landing, orbit or return action.

8.4 REQUIRED OBJECTIVES

OBJECTIVE	COMPLETION CONDITION
01 - Capture a flyby image	Player aims at the Jupiter target and captures the image.
02 - Capture the view of Earth	Player turns the camera toward Earth and captures the image.
03 - Detect interstellar space	Player listens to the plasma waves and sees the heliopause crossing.

8.5 COMPLETION SCREEN

Title: MISSION COMPLETED - INTERSTELLAR SPACE REACHED
Learning card: “Voyager 1 was launched in 1977 to study Jupiter and Saturn. It kept going, and in 2012 it became the first human-made object to enter interstellar space, carrying a Golden Record of sounds and images from Earth.”
Buttons: “Replay Mission” and “Return to Mission Selection”.

ACCURACY NOTE
This is a simplified educational recreation. Voyager 1 is a flyby probe: it never lands, orbits or returns. Its cameras were switched off after 1990, so do not show new photographs from interstellar space. The heliopause is a boundary region, not a physical wall. Do not say Voyager 1 has left the solar system entirely; it has not reached the Oort cloud. Verify every distance and signal-time value against NASA data before publication.